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Ultrastable eddy current displacement sensor working in harsh temperature environments with comprehensive self-temperature compensation

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ABSTRACT

This paper proposes a method of self-temperature compensation for eddy current sensors (ECSs) during rapid temperature changes. The temperature drift of the sensor impedance caused by the probe coil and target temperature variations was discussed and analyzed. The proposed impedance measurement method can distinguish the temperature drift caused by the sensor coil and the eddy currents in the target, and eliminate both of them in real time. A prototype ECS was manufactured and tested. The temperature coefficients of the probe's inductance and resistance caused by the target and the coil were measured by a designed setup. The results show that the temperature drift of the inductance is mainly caused by the target, and that of the resistance is mainly from the coil itself which is much larger than the inductance's drift. The prototype ECS that uses this new comprehensive self-temperature compensation method achieved an ultra-low temperature drift of 4 nm/°C without any additional hardware or temperature sensor. It can work effectively even the environment temperature changes as fast as 10 °C/h.

1. Introduction

Requirements for high performance displacement sensors increase rapidly with the development of industrial automation and micromechanical technology. Displacement sensors should be low cost, reliable, non-contact, high resolution, and immune to environmental variation. The most important high-performance, non-contact displacement sensors are eddy current, optical, and capacitive sensors [1,2]. Optical sensors are usually bulky and very expensive. Capacitive sensors can achieve extremely high resolution and stability. However, they are very sensitive to environmental variation [1,3]. Dirt, dust, humidity, oil, or any dielectric material in the measuring gap can affect the measurement result of the capacitive sensor. Therefore, capacitive sensors are rarely used in online measurement applications. Eddy current sensors (ECSs) are inherently rugged, low cost, high performance, and insensitive to environmental contaminants, which make them the usual first choice for most industrial applications [2,4,5]. ECSs measure the impedance of the sensing coil, which allows it to achieve ultra-high resolution like capacitive sensors. However, ECSs have a much larger temperature coefficient (TC) because of the high TC of the metal resistivity, which limits their application in high-precision displacement measurement systems [6]. Different kinds of methods, such as increasing working frequency, using temperature sensor to compensate drift, compensation integrated electronics [7], and differential probe structure have been developed to reduce the drift to some extent. Nabavi [8] presented an ECS interface with remarkable performance for industry applications, but it is only for the specific applications and a little complicated. In addition, 20 MHz working frequency is hardly to be realized sometimes, because the maximum working frequency of an ECS is limited by the self-resonate frequency of the sensor coil (including the cable) [4]. Wang proposed a method to reduce the TC of an ECS by two orders of magnitude through signal processing without any additional hardware [9]. However, it is difficult to obtain very good results by the presented method when the environment temperature changes rapidly.

In this paper, we present a method that can remarkably improve the stability of an ECS by using self-temperature compensation when the environment temperature changes rapidly. This method further separates the temperature drift of the sensor impedance caused by the probe coil and target temperature variations, and measures the temperature of the probe coil in real time. Thus, both the temperature drift caused by the target and probe coil

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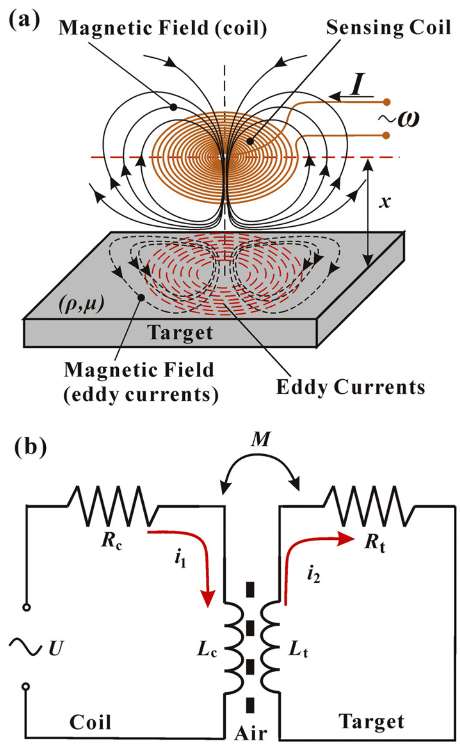

Fig. 1. Working principle and equivalent model of eddy current displacement sensor.

temperature variation can be eliminated in real time. The ECS prototype that uses this comprehensive self-temperature compensation method achieved a TC as low as 4 nm/°C even the environment temperature changes as fast as 10 °C/h (close to the performance 2.6 nm/°C when the temperature changes slowly about 1 °C/h [9]). The new method is solely based on signal processing circuits and does not require any additional hardware of the sensor coil or temperature sensor.

2. Working principle

Generally, an ECS system consists of a sensing coil and a conductive target [Fig. 1(a)]. For a non-ferromagnetic target, the basic interaction between the probe coil and the target metal can be described by the transformer model [Fig. 1(b)]. The primary of the transformer represents the sensor coil, which has a serial resistor R_c and an inductor L_c, and the secondary of the transformer represents the target, with a serial resistor R_t and an inductor L_t. According to Kirchhoff's law, the effective impedance of the ECS (input impedance of the transformer) can be obtained [10]

$$\begin{cases} R = R_c + \dfrac{\omega^2 M^2}{R_t^2 + (\omega L_t)^2} R_t = R_c + R_e \\ L = L_c - \dfrac{\omega^2 M^2}{R_t^2 + (\omega L_t)^2} L_t = L_c - L_e \end{cases} \quad (1)$$

where ω is the angular frequency of the coil's excitation, and R_e and L_e are the eddy current-caused resistance and inductance, respectively. M is the mutual inductance between the coil and the target, which depends on the distance x between the coil and the target, and can be written as $M = k\sqrt{L_c L_t}$ $(0 < k < 1)$. k is the coupling factor between the primary and the secondary of the transformer, which is a function of the distance x. The TCs of the inductance L_c and L_t mainly depend on the coefficient of thermal expansion of the coil or the equivalent model of the target. For a specific ECS system with constant geometric parameters, the inductance of the coil and the target is constant if the thermal expand is ignored. Thus, TC of the inductance is only several tens of ppm/°C and could be much lower through better manufacturing and packaging process of the coil. As copper and aluminum have the highest conductivity at room temperature, the sensor coil is usually made of copper wire and copper or aluminum is the best target material for a high-performance ECS. However, TC of the copper or aluminum's resistivity is nearly 3900 ppm/°C at room temperature. Thus, the resistance of the coil R_c and the equivalent resistance of the target R_t are very sensitive to temperature, which results in high TC of the effective impedance of the ECS. Therefore, R_c, R_t, L_c, and L_t are all functions of temperature, and k is the function of x. Thus, the effective impedance Z_{eff} can be expressed as

$$Z_{eff} = [R_c(T_c) + R_e(x, T_t)] + j\omega[L_c(T_c) - L_e(x, T_t)] \quad (2)$$

where T_c and T_t are the temperature of the coil and the target, respectively. For most high-precision applications, the temperature of the coil and the target may be the same. The temperature drift of an ECS can be completely eliminated by using the method proposed in Ref. [9]. However, this method hardly achieves effective results if the coil and the target have different temperatures when the temperature changes rapidly. For this situation, the temperature drift caused by the temperature variations of the coil and the target must be separated to achieve comprehensive and effective compensation.

As the ECS signal processing circuit directly detects the total part of the effective impedance, without distinguishing the coil part and the eddy current-caused part, the inductive part (L) and the resistive part (R) of the sensor coil [Eq. (2)] can be rewritten as

$$\begin{cases} L = f(x, T_c, T_t) \\ R = g(x, T_c, T_t) \end{cases} \quad (3)$$

According to Eq. (1), TC of the eddy current-caused part impedance (L_e and R_e) depends on the measurement distance x, which means that the TC of the ECS is not constant. Since the measurement range is usually only tens of microns and the temperature fluctuation is a few degrees for a high-resolution ECS system, the inductive part (L) and the resistive part (R) of the effective impedance [Eq. (2)] can be simply expressed as the linear combination of x, T_c and T_t

$$\begin{cases} L = K_{11}x + K_{12}T_c + K_{13}T_t + L_0 \\ R = K_{21}x + K_{22}T_c + K_{23}T_t + R_0 \end{cases} \quad (4)$$

For a specific sensor coil and target material, K_{ij} ($i = 1, 2$; $j = 1, 2, 3$) are constants and can be measured previously and used as correction parameters. Except for measuring the inductance and the resistance, one of the temperatures (T_c or T_t) also needs to be measured at the same time. Then, the true displacement information x can be obtained by eliminating the drift caused by the target and the coil temperature variations completely. Measuring the temperature of the target requires an additional temperature sensor, which results in an electronic connection to the target, whereas the coil temperature can be obtained by measuring the coil resistance, which is a better choice. The relationship between the coil resistance and its temperature can be written as

$$R_c = R_{c0} + K_c T_c \quad (5)$$

Thus, the coil temperature can be obtained by

$$T_c = (R_c - R_{c0})/K_c \quad (6)$$

Substituting Eqs. (6) into (4) and solving Eq. (4), the temperature-independent information x can be calculated. Without any additional hardware, the position information x can be simply extracted by adding the R, L, and T_c signals with different

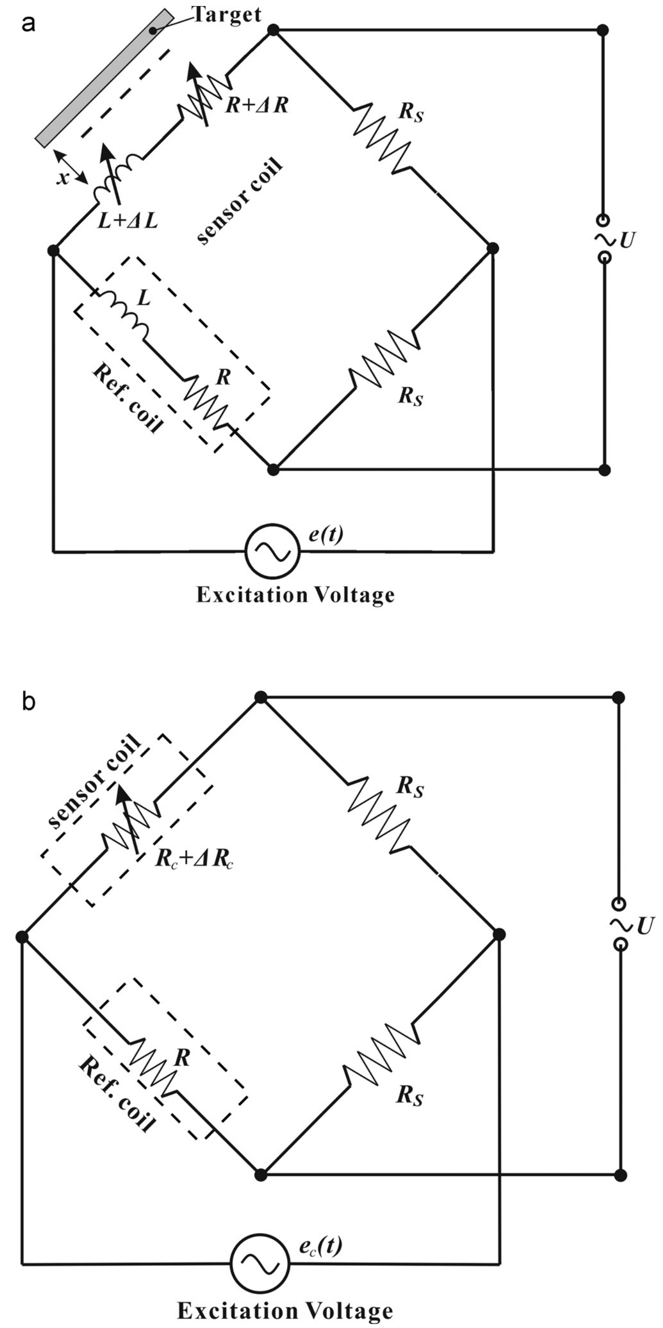

Fig. 2. AC bridge circuit of ECS: (a) for displacement measurement, (b) for coil temperature measurement.

coefficients. As a result, the influence of temperature on the position information caused by both the coil and the target is cancelled, and even the temperature of the target T_t can be obtained at the same time.

According to the analysis above, the main issue is how the R and the L parts of the coil can be separated and how the coil resistance can be measured at the same time. As Ref. [9] proposed, an AC voltage source bridge [Fig. 2(a)] was used to separate R and L, and very good results were obtained.

The value of the sampling resistor is selected as $R_s = \omega L - R$, and $\Delta R << \omega L$, $\Delta L << L$ is assumed, then the output voltage of the bridge [Fig. 2 (a)] can be simplified as

$$\tilde{U} = \frac{\tilde{e} R_s}{2\omega L}\left(-\frac{\Delta L}{L} + j\frac{\Delta R}{R_s + R}\right) \tag{7}$$

where ΔR and ΔL are the variations of the effective resistance and inductance caused by the target movement or temperature drift. According to Eq. (7), ΔR and ΔL can be separated easily by a dual-channel quadrature lock-in amplifier. For measuring the coil resistance, a low-frequency AC bridge was used because the AC bridge has a much smaller drift than the DC bridge, and the eddy current effect on the target at low frequency can be completely ignored. To simplify the measurement circuit, the low-frequency voltage signal was applied to the same bridge. The inductive reactance of the probe coil and the reference coil are only several mΩ at low frequency and they are stable with temperature. Therefore, the inductive reactance of the two coils can be ignored in this bridge. Thus, the equivalent circuit of the low frequency bridge for the coil temperature measurement is composed of four resistors, as shown in Fig. 2(b). The output voltage of this bridge can be calculated as

$$U_c = \frac{e_c R_s (R_c - R)}{(R + R_c)(R_s + R_c)} + \frac{e_c R_s}{(R + R_c)(R_s + R_c)} \cdot \Delta R_c \tag{8}$$

where e_c is the amplitude of the low-frequency voltage signal applied to the bridge. The first term in Eq. (8) is constant and can be ignored, which is caused by the difference of the sensor coil resistance R_c and the reference coil resistance R.

3. Device description

An ECS prototype was manufactured to demonstrate the analysis in the sections above. The ECS consists of a spiral planar copper coil as sensing coil, a 0.2 mm thick aluminum plate as the target, and a signal conditioning board to demodulate the impedance signal (Fig. 3). The sensor coil was wound with a 50 μm OCC wire, which consists of 30 turns and has a 5 mm outer diameter and a 2 mm inner diameter. The coil is excited with a 1 MHz sinusoidal voltage for displacement measurement, with an inductance (L_c) of 4.75 μH and a resistance (R_c) of 4.6 Ω (including coaxial cable). As shown in Fig. 3, the signal processing circuit consists of two excitation voltages (the high-frequency voltage for eddy current testing and the low-frequency voltage for coil temperature measurement), a well matched AC bridge, a preamplifier (INA103, a low noise, low distortion, instrumentation amplifier), three lock-in amplifiers (AD630, a high-precision balanced modulator/demodulator), and output amplifiers. The reference coil was wound with 0.25 mm constantan alloy wire on a quartz glass rod and configured in the signal processing circuit board. The TC of the reference coil's inductance is below 20 ppm/°C, and the TC of its resistance is about 10 ppm/°C to 20 ppm/°C. The two sample resistors (R_s) have ultra-high precision and are stable with 0.01% accuracy and only 5 ppm/°C temperature drift. In our experimental setup, the distance between the target and the probe coil is around 180 μm; at this distance, the sensing coil has an inductance (L) of 2.07 μH and a resistance (R) of 5.8 Ω. The reference coil was wound carefully to match the parameters of the coil target at the frequency of 1 MHz. As a result, the unbalanced output of the AC bridge is very small. The lock-in amplifier that uses $e(t)$ as reference signal obtains the inductance change signal U_L, and the quadrature demodulation channel obtains the resistance change signal U_R, while the lock-in amplifier that uses $e_c(t)$ as reference signal obtains the coil resistance signal U_C. The gain and the offset of the output amplifier can be adjusted according to the measuring range. Thus, if the coefficients in Eqs. (4) and (6) are known, the temperature-independent displacement signal can be easily calculated.

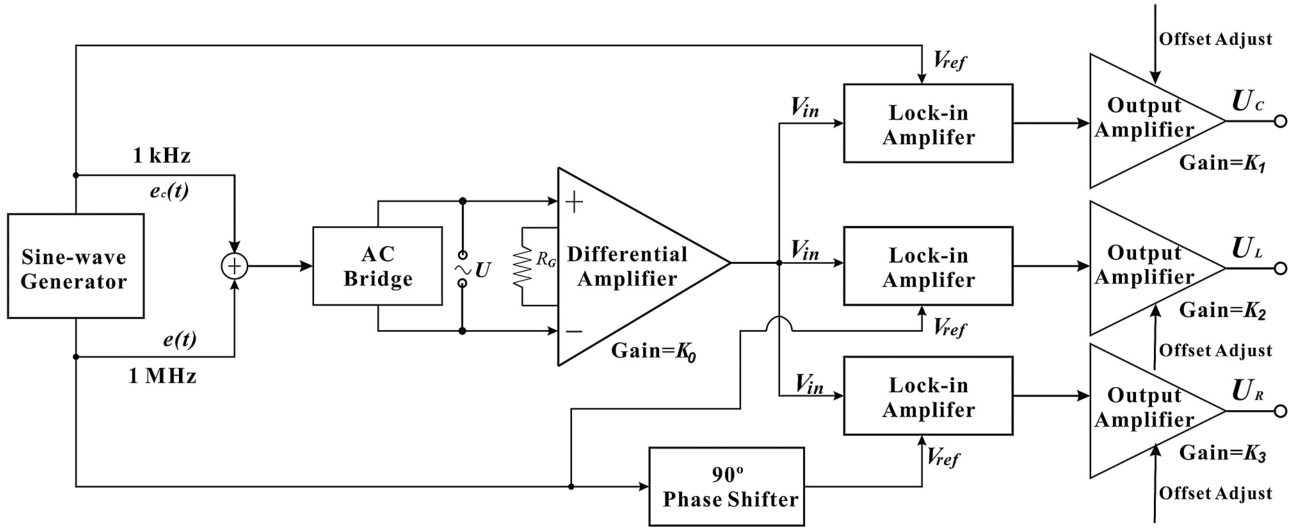

Fig. 3. Schematic of the signal processing circuit of the ECS.

Based on Eqs. (7) and (8) and the signal processing circuit (shown in Fig. 3), the three channel output voltage are

$$\begin{cases} U_C = \dfrac{e_c R_s}{(R_s + R_c)(R + R_c)} \cdot \Delta R_c \cdot K_0 \cdot K_1 \\ U_L = \dfrac{e R_s}{2\omega L} \cdot \dfrac{\Delta L}{L} \cdot K_0 \cdot K_2 \\ U_R = \dfrac{e R_s}{2\omega L} \cdot \dfrac{\Delta R}{R_s + R} \cdot K_0 \cdot K_3 \end{cases} \tag{9}$$

where e and e_c are the amplitudes of the excitation voltage, K_0 is the gain of the differential preamplifier, and K_1, K_2, and K_3 are the gain of the three output amplifiers, respectively. ΔR_c is the variation of the DC resistance of the coil, which is caused by the coil temperature variation. In our experiments, $e = 1.0\,\mathrm{V}$, $e_c = 1.0\,\mathrm{V}$, $L = 2.07\,\mu\mathrm{H}$, $R_s = 7.2\,\Omega$, $R_c = 4.6\,\Omega$, $R = 5.8\,\Omega$, $f = 1\,\mathrm{MHz}$, $\omega L = 13.0\,\Omega$, $K_0 = 61$, $K_1 = 30.09$, $K_2 = K_3 = 29.29$. Thus, the variation of inductance and resistance of the coil can be obtained by measuring the value of U_L and U_R, and calculated by using

$$\begin{cases} \sigma_L = \dfrac{\Delta L}{L} = \dfrac{U_L}{494.5} \\ \sigma_R = \dfrac{\Delta R}{R} = \dfrac{U_R}{220.6} \end{cases} \tag{10}$$

where σ_L and σ_R are the normalized variation of the coil's inductance and resistance, respectively, and are dimensionless quantities.

4. Experiments and results

First, the sensitivities of σ_L and σ_R to the displacement were measured. A single-axis piezo-nanopositioner (P-620.1CD-Linearity, Physik Instrumente GmbH & Co. KG, Germany) was used to generate a linear 10 μm displacement variation. The results show that the sensitivities of U_L and U_R to the displacement are 0.8313 and −0.2222 V/μm, respectively, or equal to the sensitivities of σ_L and σ_R to the displacement, which are 1681 and −1007 ppm/μm, respectively.

Measuring the sensitivities of σ_L and σ_R to the target temperature and coil temperature is a challenging work. First, as nearly all metallic or nonmetallic materials have a relative high thermal expansion, the distance between the coil probe and the target changes with temperature because of the frame's thermal expansion. The displacement drift from the frame's thermal expansion may be even larger than the inherent temperature drift of the sensor itself. Furthermore, heating the target (aluminum plate) or the copper coil alone and keeping the temperature of the other parts constant is nearly impossible, which prevents the measurement of the coefficients of the impedance to the target temperature and the coil temperature.

To solve the first problem, a setup [9] with very low temperature drift was specially designed [Fig. 4(a)]. A micrometer head was assembled on an aluminum frame, and the ECS probe

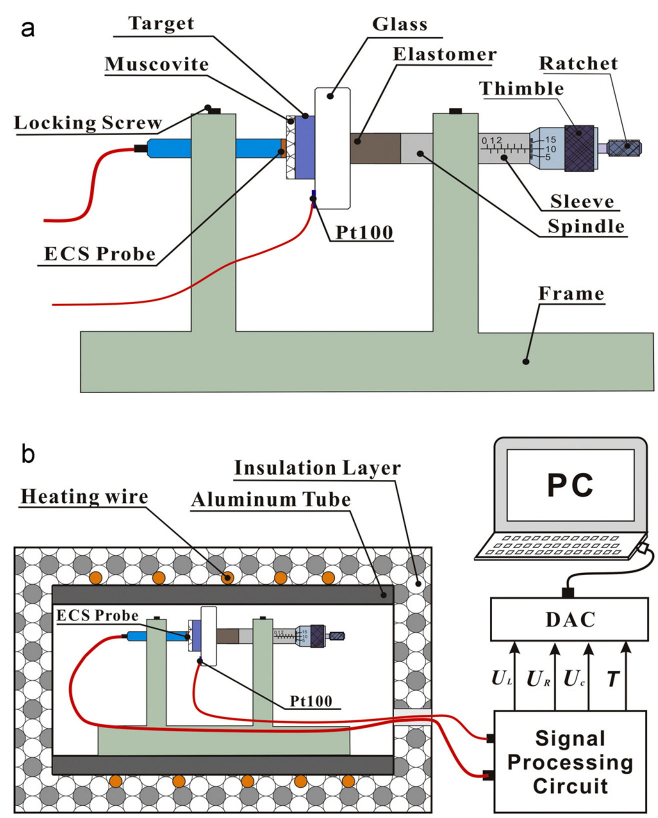

Fig. 4. Temperature drift measurement system (a) the structure with fixed distance between the coil and target; (b) the configuration of the temperature drift measurement system equipment.

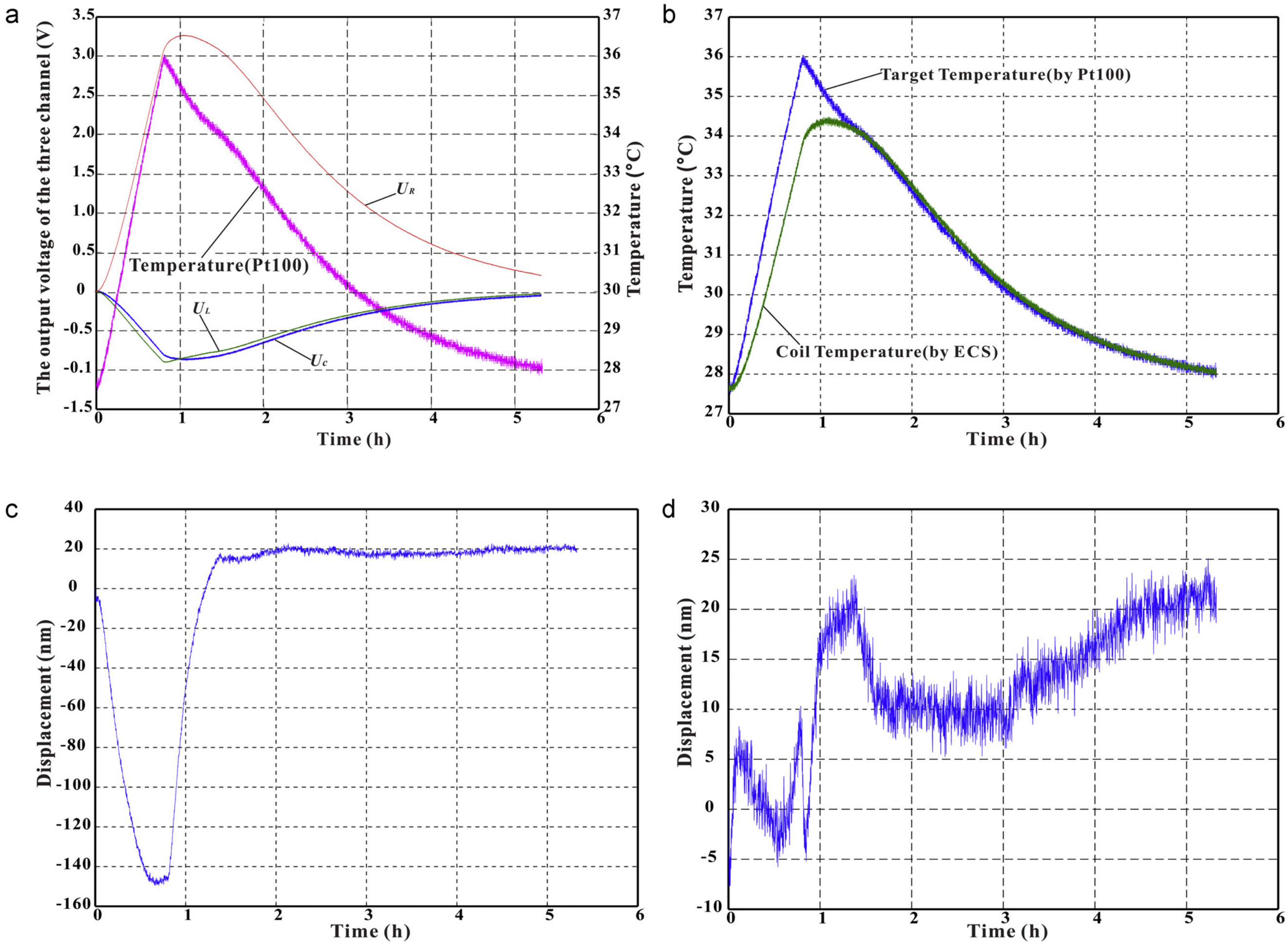

Fig. 5. ECS probe's temperature drift and compensation results during heating and cooling (a) the change of the three output channel and the temperature within the process. (b) The target temperature and the coil temperature with time. (c) The displacement output of the ECS that uses the old self-correction method within the process. (d) The displacement output of the ECS that uses the comprehensive compensation within the process.

was assembled coaxially with the spindle of the micrometer. A 0.2 mm thick aluminum plate was selected as the target plate. A 180 μm thick muscovite plate was used to determine the distance between the coil and the target surface. When the micrometer screw spindle moved to the left to compress the elastomer at a suitable force, the distance between the coil on the probe and the target surface was fixed to the thickness of the muscovite plate. The muscovite plate and the probe maintained contact, and the thermal expansion of the muscovite plate's thickness is only 1.6 nm/°C (with a general thermal expansion coefficient of 9 ppm/°C for muscovite). The thickness variation caused by thermal stress is also negligible as the stiffness of the muscovite plate is much greater than that of the elastomer. Therefore, the distance between the target surface and the coil on the probe can be regarded as constant during the entire measuring process. The setup was placed in an aluminum tube surrounded by a thermal isolation box [Fig. 4(b)]. A heating wire wound on the aluminum tube was used to generate thermal energy and induce a uniform temperature increase of the surrounding environment. A Pt100 was placed near the sensing coil to measure the temperature inside the aluminum tube. This temperature was used to calculate the thermal sensitivity of the coil impedance. The test setup was heated and cooled slowly to ensure constant temperature throughout the inner space of the aluminum tube. The well designed circuit was placed in an air conditioned room and regarded as stable.

For the second problem, the TC of the coil impedance was measured first. The probe coil without a target was placed in the thermal isolation box and heated. A new well matched bridge [similar to the one shown in Fig. 2(a)] was used to measure the TC of the coil's inductance (m_L) and resistance (m_R); the obtained TCs are 12.3 and 3133 ppm/°C, respectively. Then, the total (effective) impedance's TC was measured by heating the whole coil-target unit with a fixed distance simultaneously and slowly. The TC of the resistive part (q) is 2292 ppm/°C, and the TC of the inductive part (p) is 255.6 ppm/°C.

Table 1
Parameters of impedance and temperature coefficients.

Parameters	L	R	L_c	R_c	L_e	R_e
Values	2.07 μH	5.8 Ω	4.75 μH	4.6 Ω	2.68 μH	1.2 Ω
Coefficients	p	q	m_L	m_R	n_L	n_R
Values (ppm/°C)	255.6	2292	12.3	3133	−173.8	−931.8

Thus, the TC of the effective impedance caused by the temperature drift of the coil part and that of the eddy current part can be written as

$$\begin{cases} \Delta L = pL \cdot T = p(L_c - L_e) \cdot T = (m_L L_c + n_L L_e) \cdot T \\ \Delta R = qR \cdot T = q(R_c + R_e) \cdot T = (m_R R_c + n_R R_e) \cdot T \end{cases} \quad (11)$$

where p, q, m_L, m_R are known, n_L and n_R are the TC of the eddy current caused inductance and resistance, which can be calculated. The parameters and the TC of the resistance and inductance of the coil are listed in Table 1.

The ECS signal processing circuit directly detects the total part of the effective impedance without distinguishing the coil part and the eddy current caused part. When the temperatures of the target and the coil change differently, the temperature drift of the impedance can be written as

$$\begin{cases} \Delta L = p_c L \cdot T_c + p_e L \cdot T_t = m_L L_c \cdot T_c - n_L L_e \cdot T_t \\ \Delta R = q_c R \cdot T_c + q_e R \cdot T_t = m_R R_c \cdot T_c + n_R R_e \cdot T_t \end{cases} \quad (12)$$

where p_c and p_e are the TC to the total inductance caused by the coil and the target temperature variations, respectively, and q_c and q_e are the coefficients to total resistance. According to Table 1, p_c, q_c, p_e, and q_e can be easily calculated. In summary, the coefficients in Eq. (4) are all known, and Eq. (4) can be rewritten as

$$\begin{cases} \sigma_L = d_L x + p_c T_c + p_e T_t + \sigma_{L0} = 1681x + 28.22T_c + 225.0T_t + \sigma_{L0} \\ \sigma_R = d_R x + q_c T_c + q_e T_t + \sigma_{R0} = -1007x + 2485T_c - 192.8T_t + \sigma_{R0} \end{cases} \quad (13)$$

where the unit of σ_L, σ_R, σ_{L0} and σ_{R0} are ppm, μm for x, and °C for T_c and T_t.*σ_{L0} and σ_{R0} are constant.*

For simplifying the correction, substituting Eqs. (10) into (13) yields

$$\begin{cases} U_L = 0.8313x + 0.0139T_c + 0.1113T_t + U_{L0} \\ U_R = -0.2222x + 0.5482T_c - 0.0425T_t + U_{R0} \end{cases} \quad (14)$$

As an additional result of the experiment above, the relationship between the coil temperature T_c and the output voltage U_C is

$$T_c = 27.6 - 7.6 \times U_C \quad (15)$$

Substituting Eqs. (15) into (14) and solving the equation yields

$$\begin{cases} x = 0.7069U_L - 1.856U_R - 7.659U_C + x_0 \\ T_t = 13.73U_L + 3.670U_R + 57.57U_C + T_{t0} \end{cases} \quad (16)$$

where the unit of the three voltages is Volt(V), μm for x, and °C for T_t. x_0 and T_{t0} are constants that can be adjusted by the offset nulling of the three output amplifiers according to the target position and temperature point of the ECS probe.

To verify the effect of this temperature compensation method, the ECS with a fixed distance between the target and the coil was placed in the thermal isolation box, which was then heated rapidly and cooled naturally. The voltage changes of the three output channel and the temperature variation during the heating and cooling process were shown in Fig. 5(a). The temperature curve shows that the temperature rose rapidly by 8.5 °C within about 50 min and then came down slowly in the next 4.5 h. The coil temperature is calculated from the voltage U_C and shown in Fig. 5(b) with the temperature of the target (measured by Pt100) during the process. The curves show that the temperature of the probe coil changes more slowly than that of the target, and the coil temperature does not remain the same as the target temperature when the environment temperature changes too fast. Fig. 5(c) shows the displacement output of the ECS that uses the old self-correction method proposed in Ref. [9]. The maximum error is about 150 nm, which appears at the point where the coil and the target have the maximum temperature difference. Furthermore, the curve trend of U_L [Fig. 5(a)] is similar to the temperature curve of the target, while the curve trend of U_R is similar to the temperature curve of the coil, which is consistent with the temperature coefficients in Eq. (16). The displacement output of the ECS obtained by using the new self-correction method is shown in Fig. 5(d), which shows that the total drift during the process is lower than 30 nm. Hence, the temperature drift of an ECS that uses the proposed temperature compensation method is below 4 nm/°C in an environment where the temperature changes at a speed of about 10 °C/h.

5. Conclusion

The results prove the validity of the method presented in the second section. Separating the temperature drift caused by the target temperature variation and the coil temperature variation, and measuring the coil temperature in real time can eliminate the temperature drift of an ECS nearly completely. The results indicate that an ECS with this proposed self-correction method can achieve an ultra-low temperature drift of 4 nm/°C even in an environment where the temperature changes rapidly; this temperature drift is only one-fifth of the TC obtained by using the old self-correction method under the same conditions. This method ensures that a high-resolution ECS remains very stable even in a harsh temperature environment. As an additional result, the measured coefficients show that the temperature drift of the inductance part is mainly from the target, while that of the resistance part is much larger and mainly from the coil itself. These findings provide some guidance for designing a high-performance ECS, especially high-resolution and low-temperature drift ECS. The residual drift after the correction may come from several aspects, such as the temperature drift of the probe cable's resistance, the error of the correction coefficient, the variation of the TC of the ECS at different temperatures, and the temperature drift of the signal processing circuit. A relative long probe cable is needed for universal displacement sensor to adopt different kinds of applications, but long cable will reduce the temperature stability and performance of an ECS. Using this method, the cable resistance variation has no influence to the thermal stability, since the cable resistance can be regarded as part of the coil resistance. Generally, the cable and coil resistance have similar temperature variations, which is a common situation according to the coil packaging. When the temperatures of the coil and the cable change differently, the 4-wire measurement method could be adopted.

This paper presents a method that can remarkably improve the stability of an ECS by using self-temperature compensation when the environment temperature changes rapidly. The impact factor of the temperature drift of the ECS was discussed and analyzed, and the temperature coefficients caused by the coil and the target were measured. The ECS that uses this comprehensive compensation method is mainly based on the signal processing of the sensing coil. Thus, the ECS is low cost and has a simple structure. The high-resolution ECS that uses the proposed compensation method will have competitive advantages in most industrial applications, especially when the environment temperature is not well controlled, such as the hot metal production line in the factory, or the precision equipment in the wild. This method can be directly used to non-contact metal temperature measurement, since it can measure the displacement, the coil temperature and the target temperature at the same time with only one detection coil. In addition, this techniques could be used for any impedance measurement based eddy current testing applications.

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Biographies

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